Cognitive Traits Across Domains: Classification, Symmetry and the Cost of Interaction

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Abstract

A person asked about focus, organization and initiation, who is late with the laundry and, by his account, holds a large network of considerations in mind while building a theory, understands the expected answer to be that his executive functioning is poor. The paper starts from that case and asks what licenses moving from performance in a sampled domain to a general attribution of a function. It distinguishes functions from their manifestations, descriptions from attributions, and both from evaluations and from the assistance a person wants; it represents a classification scheme by its sampled domains, its indicator rules and its designated reference branch; and it reads the Principle of Symmetry as a claim about admissible redrawings of such schemes. Two results are proved: each of two branches can be given, on an inventory selected for it, a non-empty list of lower traits only exactly when each is higher than the other on some competence by more than the measurement's margin; and the descriptions of a reference and another branch trade places literally when every function's comparison reverses sign between two sampled domains. Whether that paired crossing holds for autistic and non-autistic branches on independently defensible domains is stated as an empirical conjecture, with what would count against it; a counterexample shows that it is not a theorem about schemes in general. A cost model is developed separately: the effort an interaction extracts is a baseline plus a mismatch between a person and the conventions the interaction runs on, to be tested in a within-dyad crossover designed here, whose simulation is reported in a supplement.

1 Introduction

1.1 A case

A clinician asks how a person's focus, organization and initiation are, the three things her questions gather under executive functioning. The person is late with the laundry, and understands that the expected answer is "not great". The same person, by his own account, holds a large network of considerations in mind at once while building a theory by inductive reasoning. The case is the author's (Appendix B), and the paper takes from it one question, which it treats as the central one: what warrants letting performance in one domain determine whether a person possesses a

*Email: akb9408@rit.edu, Website: `https://aethus.org`. This paper develops, as a self-contained argument about classification and a proposal in the psychology of communication, two principles stated in a larger framework [1]: the Branch Principle and the Principle of Symmetry, with the observation that a difference which switches with context refutes an unconditional claim of incapacity. The framework's metaphysics plays no role in the argument; what is carried over is one piece of bookkeeping, the treatment of a diagnostic label as a partial information state, explained where it enters.

function in general? On his reading, the questions sampled a domain, domestic activity, and the description they invited was of the person. Sampling one domain does not, by itself, warrant a general attribution of the possession or absence of a function. An assessment can earn a broader scope, but it earns it through an independently justified account of how its sampled manifestations generalize, and the paper asks what such an account has to contain.

Three distinctions do the work of the paper (Section 3). A *function* is distinguished from its *manifestations*: an instrument that samples focus in a household task has not measured focus in sustaining a question across competing explanations. A *description* of what someone does is distinguished from an *attribution* of the processes involved: that a person misses deadlines is a description, that a person lacks organization an attribution, and the step between them is an inference whose warrant is the question just asked. And both are distinguished from an *evaluation* relative to a goal and from the *assistance* a person wants, which remain real whatever the attribution: the laundry is late, and that is information the person may want acted on. The author's account of his case, that his executive functioning has been picked up and moved into theory-building, is an explanatory hypothesis the contrast motivates and does not establish; and the clinician's questions, his reading of what they invited, and claims about any standardized instrument, which the conversation did not involve, are kept apart.

1.2 The argument, and what is established

The paper develops two principles the author stated in a larger work [1], quoted with their dates, elisions marked, in Appendix B; the formal apparatus, the cost model and the design are this paper's. The *Branch Principle* holds that "ontologically neurodivergent people are better understood as a collection of categorical branches and 'allistic' is a single branch, such that two random branches are typically as different from one another as they each are to a third". The *Principle of Symmetry* holds that "you can always redraw the conventional category boundaries in psychology so that allistic and a given neurodivergent branch 'trade places phenomenologically'". The paper reads the second as asserting that there is, for the reference branch a classification scheme designates and a given other branch, an admissible redrawing of the scheme under which their descriptions trade places, a scheme being the domains it samples, the indicators it uses for each function and its reference, and the remainder of the branches being carried along with the designation. "Admissible" is where the content lies, since with unrestricted reassignment of indicators the trading of places holds by definition (Section 3.2).

What the classification argument proves, what it conjectures and what it does not claim are set out in Section 3.4. Of the source's further claims, the remark that autistic people "are described as taking a script to allistic conversations", when "if the shoe were on the other foot" non-autistic people would be, is the seed of the paper's second part.

1.3 A separately testable development

If descriptions are scheme-relative, the difficulty of an interaction between differently situated people can be asked about in the relation between a person and the conventions the interaction runs on, rather than in the person. The double empathy problem [2] already relocated the difficulty from the autistic person to the pair, a mutual failure of understanding between people whose ways of experiencing and communicating differ, with a decade of empirical work behind it [3, 4, 5]; Milton's account also has the conventions in view, social subtext being actively constructed and a label that locates the problem in the other resolving the one who applies it. What the accounts examined here do not have is the conventions as a measured and manipulable variable

with a prediction of how a change in them redistributes effort between the partners. The paper's second part supplies that as a hypothesis independent of the first, which motivates and does not entail it: a cost model that predicts each participant's change in effort from profiles measured and norms coded in advance, to be tested in a within-dyad crossover designed here (Section 5 to Section 7). Aptitude-by-treatment interactions are its theoretical setting [6, 7], with the aptitude a premeasured profile rather than a trait inferred from the outcome and the treatment a norm imposed on both partners at once; its nearest rival is the double-empathy literature's formal model of channel preference [8] (Section 6).

Two commitments govern the writing: the scope rule of Section 2 for every claim about "the autistic", and the recommendations to autism researchers on ableist language [9], so that "autistic" and "non-autistic" are the groups' names, "allistic" appears only inside the source's quotations, and no competence difference is called a deficit, "deficit" and "impairment" appearing only where the clinical literature's own usage is reported.

1.4 Plan

Section 2 fixes what a label licenses, Section 3 is the classification argument and Section 4 says what a difference that switches with context shows. Sections 5 to 8 develop the cost model and its test, the simulation in summary, the supplement carrying the full study; Sections 9 to 11 give what published results constrain, what lies outside the paper, and the discussion. Appendix A lists the preregistration items and Appendix B the source passages.

2 A label as a partial information state

A diagnostic label carries what it records and no more. The larger framework calls such a state an Aethus and any fuller description of the same person a proper child of it [1]; this paper says *partial information state* and *refinement*. The point of the vocabulary is not metaphysical. It is that a label is an object with a definite content, and that a claim about everyone who bears it is licensed by that content or not at all.

Three kinds of claim about the bearers of a label are therefore distinguished.

(i) *Definitional entailment.* What the label records holds of every bearer. The diagnostic criteria for autism spectrum disorder in the fifth edition of the *Diagnostic and Statistical Manual* require all three of the social-communication features and at least two of the four restricted and repetitive features, currently or by history [10]. So "every bearer shows at least two of the four" is entailed. "Every bearer has sensory differences" is not: sensory reactivity is one of the four, and the label does not record which two.

(ii) *Group-level statistical prediction.* A claim about the distribution of bearers. It is true or false of the group, and attributing it to each member is the ecological fallacy [11].

(iii) *Individual attribution.* A claim about a particular bearer, which needs information about that bearer.

Definition 1 (Scope rule)**.** A claim about every bearer of a label is licensed in two ways only: the label's definition entails it, or independent evidence establishes it for every bearer. A claim that needs information the label does not carry is made of a refinement that carries it, or not made.

The rule does not forbid universal claims; it says where they come from.

Table 1: The executive-functioning case: three functions, and candidate manifestations of each in the domain the clinician's questions sampled and in the domain the author's account names.

Function attributed	Candidate manifestation in domestic activity	Candidate manifestation in theoretical work
Focus	maintaining attention through a household task	sustaining a question across competing explanations
Organization	coordinating steps, materials and deadlines	coordinating premises, dependencies and unresolved problems
Initiation	beginning an intended practical task	beginning an investigation, or a new argument

The source's illustration. The source gives one instance of a claim that violates the rule (Appendix B): for any trait some autistic branch can be found on which it presents as infantile, which seems to confirm that "autistic people are infantile in this way". With $I(b, \tau)$ for "τ presents as infantile on b relative to a chosen reference branch" and $\mathcal{B}$ the branches the label collects, the step $\exists b \in \mathcal{B}\ I(b, \tau) \not\Rightarrow \forall b \in \mathcal{B}\ I(b, \tau)$ violates the scope rule, and the interchange of $\forall\tau\,\exists b$ with $\exists b\,\forall\tau$ makes the collected witnesses look like one profile; the confirmations depend on the partition chosen, the reference class problem [12] in categorical form.

3 Classification schemes and the Principle of Symmetry

3.1 The executive-functioning case, taken apart

The case of Section 1 has a family of functions, executive functioning, three functions the clinician's questions gather under it, focus, organization and initiation, and two domains in which each might show. Table 1 sets the domains side by side. Both columns list candidate manifestations, the right-hand one activities that would count as the function's showing in theoretical work if the correspondence between the columns holds, not activities already shown to engage the same processes as the left-hand ones.

The inference the case questions runs from the left-hand column to the person: performance in the sampled domain is taken to settle whether the person has the function. Its warrant is the assumption that the sampled manifestations are representative of the function's manifestations in general, and that assumption is what the right-hand column puts in doubt. The research vocabulary does not exclude the right-hand column: reasoning, problem-solving and planning are counted among the higher-order executive functions in the standard review [13], so the recognition of executive activity in theoretical work is already available to the field. And the measurement literature has a finding that motivates the question without answering it: across twenty studies, performance-based measures of executive function and ratings of it correlated with a median of .19, and their reviewers concluded that the two kinds of measure capture different constructs or levels of cognition [14]. The finding does not show that the two are manifestations of one common function, nor that their divergence arises from the domain sampling at issue here, and it establishes no symmetry between branches. What it shows is that indicators of one nominal function need not be interchangeable, which is reason to examine the relation between indicators and the functions they are taken to attribute; Table 1 and the admissibility conditions of Section 3.2 articulate that examination.

The objection the case makes is therefore not that everyone has strengths and weaknesses, nor that the field has no visibility into reasoning or context. It is a claim about the relation among a concept, its selected indicators and the person described through them: an assessment that samples some manifestations and reports a general attribution, without an independently justified account of how they generalize, has drawn a conclusion whose generality it has not justified. The author's historical and political objections to the deficit reading of the categories motivate the project and are not argued here; the measurement argument stands without them.

3.2 Classification schemes and admissible redrawings

The descriptive content of a diagnosis is a description assigned by a scheme, and the scheme can be written down.

Definition 2 (Classification scheme)**.** A classification scheme Σ consists of a finite set of branches $\mathcal{B}$ with a designated reference branch $r_\Sigma \in \mathcal{B}$, whose complement $\mathcal{B} \setminus \{r_\Sigma\}$ is the remainder the scheme's labels collect; a set Φ of functions it attributes; for each $f \in \Phi$ a sampled domain $\Delta_\Sigma(f)$ of activities and an indicator rule that maps a branch's performance in $\Delta_\Sigma(f)$ to a trait description, a rule that may or may not read the performance against r_Σ at a stated margin; and the category boundaries the labels draw on $\mathcal{B}$. The description $F_\Sigma(b)$ is the list of trait descriptions the scheme assigns to branch b over Φ.

Whether a scheme's traits are comparative in disguise, whether "poor organization" is "lower than the reference on organization in the sampled domain", is a question about its indicator rules, answered scheme by scheme; the comparative result of Section 3.3 treats the rules that are, and nothing here requires that every trait be. A *redrawing* of Σ is a scheme Σ' together with a bijection π of $\mathcal{B}$ that exchanges r_Σ with a branch b, with $r_{\Sigma'} = b$, and so carries the remainder $\mathcal{B} \setminus \{r_\Sigma\}$ to $\mathcal{B} \setminus \{b\}$; this is the structure the source insists on, that the exchange is between a single branch and the collection of the remaining ones, never between "allistic" and "autistic" as if the second were a branch, which the Branch Principle counts a category error (Appendix B). The redrawing *trades places* for the pair when

$$F_{\Sigma'}(\pi(x)) \;=\; F_\Sigma(x) \qquad \text{for } x \in \{r_\Sigma, b\}, \tag{1}$$

the description the old scheme gave the reference being the one the new scheme gives b, and conversely. Equation (1) is satisfiable by definition if Σ' may reassign indicators at will, and it then says nothing about cognition. The content lies in what redrawings are admitted.

Definition 3 (Admissible redrawing)**.** A redrawing (Σ', π) of Σ is admissible when

(A1) it attributes the same functions, $\Phi_{\Sigma'} = \Phi_\Sigma$;

(A2) for each function f the new sampled domain $\Delta_{\Sigma'}(f)$ consists of activities that are manifestations of f under a correspondence stated independently of the outcome, which says why an activity in the new domain and one in the old are the same function at work, what differences between them are held not to matter, and on what evidence;

(A3) the indicator rules are applied with the standards of the old scheme, the same margin and the same reliability requirements, read against the new reference.

A worked application of (A2): organization. Under (A2) each row of Table 1 is a proposed correspondence, and what it takes to make one good can be shown for organization. The proposed common process is the maintenance of dependencies among subgoals: holding which steps wait on

which, keeping the unfinished ones available while others proceed, and returning to each when what it waits on is done. Domestic activity gives evidence of it when a task has that structure, the laundry's wash, dry and fold with their waiting intervals; theoretical work gives evidence of it when a reasoner tracks which claims wait on which lemma, keeps open problems available while others are pursued, and returns to them. The correspondence is then a claim that the two are one process, and it is defensible only with the differences between the domains named and held apart: familiarity, the household routine being practised and the theory new; motivation and interest, which may run the other way; external reminders, the machine's signal against the reasoner's notes; time pressure, fixed in one and self-paced in the other; and task structure, the household's dependencies set by the world and the theory's generated by the reasoner. Three kinds of evidence bear on it, none sufficient alone. A task analysis states each task's dependency structure, the pending subgoals, the delays and whether a return is signalled, and matches the two tasks on it; signals and open-endedness matter, since three patients with prefrontal injuries who scored well on tests of intelligence failed open-ended tasks with several subgoals, a failure their authors traced to an inability to reactivate intentions after a delay without a signal [15], so a closed and cued reasoning task cannot be assumed to stand in for theory-building. Varied structure tests the correspondence. Existing tasks vary it in one setting each, a simulated breakfast of foods with different cooking times, to be finished together, run with the cooking foods in view or out of it [16], and matrix problems, closed and cued, whose error rates rise across items with the number of rules they contain [17]; a study varies dependency load in both settings, one process predicting changes in the same direction within each group. A common factor with loadings equal across groups shows that the indicators relate to it to a similar degree [18], not that the tasks instantiate one process; it is one contribution, and the argument commits to no factor model.

What is compared. Crossed scores on particular tasks, crossed performance in the domains the tasks represent, and a difference in the latent function itself are different conclusions; Conjecture 1 below asserts the second, which indicator rules read and tasks support only as far as they represent their domains. On a one-factor model the third cannot cross, each group having one factor mean: a crossing lies in the indicators' intercepts, and the factor-mean difference is identified only by an anchor intercept. Intercept differences may lie in how the function shows in each domain, in scoring, or in demands the model omits, and one omitted demand has independent evidence behind it: autistic children differed most from matched controls on the most open-ended executive tasks, which their authors suggested might reflect the experimenter's implicit demands [19]; the settings are therefore matched, or varied, in whose demands are implicit, the household's being set by others and a theory's by the reasoner. A study scores both settings on one metric fixed by the dependency structure, the proportion of pending subgoals resumed within a preregistered window after what each waits on is done; measures familiarity, motivation, fatigue and the settings' sensory and social demands; and varies load, a difference in how the function shows being an interaction of group, setting and load, where an offset need not change with load. Load dependence is necessary, not sufficient: fatigue, low motivation and sensory load also bite hardest at high load, and a bounded proportion makes an offset interact with load at floor and ceiling, so the interaction is tested on the logit scale, with the nuisance measures as moderators of the load effect; and interest in a domain may be part of how the function shows, which the design records without settling. A difference that fails these tests keeps its interpretation open. Establishing the correspondence is not establishing a reversed comparison: two activities can be organization in the same sense and give no crossed performance between two branches; the correspondence makes a redrawing admissible, and only the comparison measured in the new domain decides whether it trades places.

The universal form is not a theorem. Read over every scheme the definitions permit, the claim that for a designated branch and any other an admissible redrawing trades places is false, and the reason is instructive. Take one function whose admissible domains are two, a margin of 1 and the comparative indicator rule kept fixed:

	reference branch r	other branch b
domain 0	8	4
domain 1	7	3

With r as reference, b receives the lower description in either domain; with b as reference, r receives the higher description in either domain; no choice of domain gives r the description b had. Admissibility constrains which domains count; it does not supply one in which a comparison reverses. The source's principle is therefore read in two forms, which the paper keeps apart: a mathematical result for schemes that satisfy a structural condition, the paired crossing of Corollary 1 below, and an empirical conjecture about particular branches, functions and domains.

Conjecture 1 (The Principle of Symmetry, empirical form)**.** Let Σ be a current clinical scheme with the non-autistic branch r as its reference, and let b be any autistic branch in the sense of the Branch Principle, identified as a class by the latent-structure analysis of Section 10, the two compared by their mean profiles on instruments held to Σ's standards. Let $\Phi^- \subseteq \Phi$ be the functions on which Σ, sampling $\Delta_\Sigma(f)$, gives b the lower description, the executive functions among them. For every $f \in \Phi^-$ there is a domain, admissible as a domain of f under (A2), in which b is higher than r by more than Σ's margin.

If the conjecture holds, Corollary 1, applied to Σ restricted to Φ^- with domain 0 the sampled $\Delta_\Sigma(f)$ and domain 1 the reversing domain, shows that the redrawing which samples the reversing domains, with b designated and the remainder carried along, trades places for the pair: the old scheme gave b the label f^- for every $f \in \Phi^-$ and r none, and the redrawn one gives r the same labels and b none. In a study the partition into branches is fixed before the domain comparisons, on independent or held-out data and on measures other than the domain scores compared, so that choosing it and exhibiting its crossed profiles are not one fitting exercise. What would count against the conjecture is a function $f \in \Phi^-$ for which, over a battery of domains each established as admissible for f, b is higher than r by more than the margin in none, at a reliability the scheme itself accepts. Since the conjecture is existential over domains, no finite battery refutes it outright; the absence of a reversal across a broad battery chosen independently of the hypothesis is evidence against it, and a theory of f's domains showing that the battery exhausts them would refute it. The executive-functioning case is the conjecture's first instance, not its evidence.

3.3 Profiles, references and trait lists

Let T be a finite inventory of competences and let each branch b have a profile $x_b \colon T \to \mathbb{R}$, a score on each competence on a common scale. The scale carries a measurement margin $\delta \geq 0$, the smallest difference the instrument resolves or the analyst is prepared to call a difference.

Definition 4 (Comparative traits and trait lists)**.** For each competence τ there are two comparative traits, τ^-, *lower on τ than the reference*, and τ^+, *higher on τ than the reference*. Relative to a reference branch r at margin δ, branch b has the trait τ^- when $x_b(\tau) < x_r(\tau) - \delta$ and the trait τ^+ when $x_b(\tau) > x_r(\tau) + \delta$; write

$$D_r(b;\delta) \;=\; \{\, \tau \in T : x_b(\tau) < x_r(\tau) - \delta \,\}, \qquad U_r(b;\delta) \;=\; \{\, \tau \in T : x_b(\tau) > x_r(\tau) + \delta \,\}$$

for the competences on which b is lower and higher than r. The *trait list* of b relative to r on an inventory $S \subseteq T$ is $L_r(b;S) = \{\tau^- : \tau \in S \cap D_r(b;\delta)\} \cup \{\tau^+ : \tau \in S \cap U_r(b;\delta)\}$.

A trait list is what a diagnosis attributes on a fixed inventory, once the evaluation is stripped from it; "deficit" is the reference's reading of τ^-, and nothing below uses it. The inventory is the scheme's choice of what to sample (Section 3.2): a competence here may be a function in a domain, focus in domestic activity and focus in theoretical work being two competences, so that a redrawing of the sampled domains is a change of inventory. Three statements follow, of three different kinds.

(S1) Reference-relativity, a definition. A comparative trait is a three-place relation among a branch, a reference and a competence, and $L_r(b;S)$ has no reference-free extension: there is no comparative list of this kind until r is fixed. The claim is about comparative traits, not every description, since "uses written reminders" names no reference. The identity $D_r(b;\delta) = U_b(r;\delta)$ holds for any two profiles, and that sign reversal is not the reversal the Principle of Symmetry asserts: a branch higher on every competence receives only higher traits however the reference is set. Choosing another reference makes another comparison, not a correction of the first, "the reference-relativity being the crux rather than a modeling convenience", in the source's formalization.

(S2) Reversibility at margin δ, and literal trading of places.

Proposition 1 (Reversibility of trait lists)**.** *For branches b and r and margin $\delta \geq 0$, the following are equivalent:*

(a) there are inventories S and S' on which b's list relative to r consists of lower traits only, $L_r(b;S) = \{\tau^- : \tau \in S\} \neq \emptyset$, and r's list relative to b consists of lower traits only, $L_b(r;S') = \{\sigma^- : \sigma \in S'\} \neq \emptyset$;

(b) there are competences τ and σ with $x_b(\tau) - x_r(\tau) > \delta$ and $x_r(\sigma) - x_b(\sigma) > \delta$.

Call (b) δ-mutual advantage. *Whether or not it holds, for any $S^- \subseteq D_r(b;\delta)$ and $S^+ \subseteq U_r(b;\delta)$ the inventory $S^- \cup S^+$ gives b the list $\{\tau^- : \tau \in S^-\} \cup \{\tau^+ : \tau \in S^+\}$ and r the mirror list; what mutual advantage adds is that both $D_r(b;\delta)$ and $U_r(b;\delta)$ are non-empty, so that each branch can be given a list of lower traits only. The number of lower traits b can be given is at most $|D_r(b;\delta)|$ and the number of higher traits at most $|U_r(b;\delta)|$; a construction with three of each needs both sets to hold at least three competences, and a finite inventory supports no finer reversal than its own size.*

Proof. (a) says that $D_r(b;\delta)$ and $D_b(r;\delta)$ are both non-empty, and $D_b(r;\delta) = U_r(b;\delta)$. $\sigma \in D_r(b;\delta)$ is $x_r(\sigma) - x_b(\sigma) > \delta$, and $\tau \in U_r(b;\delta)$ is $x_b(\tau) - x_r(\tau) > \delta$, which is (b). The remaining clauses are the definition of the lists read on $S^- \cup S^+$, with $L_b(r;S)$ the sign-reversed image of $L_r(b;S)$ for every S. □

The author's six traits are the case $|S^-| = |S^+| = 3$ of Proposition 1, available when each set holds three competences. The proposition compares two profiles and says nothing of the other branches; the reference and the remainder it leaves are the scheme's (Section 3.2). Two remarks fix its premise.

- *Non-domination is not enough above $\delta = 0$.* At $\delta = 0$, (b) is the statement that neither profile is at or above the other on every competence. At a positive margin it is stronger. Take $T = \{\tau_1, \tau_2\}$, $x_b = (1.05, 1)$, $x_r = (1, 1.05)$ and $\delta = 0.1$: neither profile dominates, yet $D_r(b;\delta)$ and $U_r(b;\delta)$ are both empty, and no inventory attributes a trait to either.

- *What is testable.* Whether a non-reference branch exceeds the reference somewhere by more than the margin is a question about measured profiles, and it is answered competence by competence. The source's own case, two competences at opposite developmental ages (Appendix B), illustrates the premise's shape; it is motivation, not a measurement on commensurate scales.

Corollary 1 (Literal trading of places)**.** *Let each function $f \in \Phi$ be scored in two domains, so that $(f, 0)$ and $(f, 1)$ are competences, and let Σ sample domain 0 with reference r and Σ' sample domain 1 with reference b, both reading comparisons at margin δ, the reference branch receiving the empty list. If for every f the comparison of b with r reverses sign between the domains, b lower than r by more than δ in one and higher by more than δ in the other, then $F_{\Sigma'}(\pi(x)) = F_{\Sigma}(x)$ for $x \in \{r, b\}$: the descriptions trade places literally.*

Proof. $F_\Sigma(r) = \emptyset = F_{\Sigma'}(b)$, the reference receiving the empty list in each scheme. For each f, $F_\Sigma(b)$ holds f^- exactly when $x_b(f, 0) < x_r(f, 0) - \delta$, and $F_{\Sigma'}(r)$ holds f^- exactly when $x_r(f, 1) < x_b(f, 1) - \delta$, which under the sign reversal is the same condition; likewise for f^+. So $F_{\Sigma'}(r) = F_\Sigma(b)$. □

The corollary's condition is stronger than δ-mutual advantage: it asks for a crossed comparison within every function, across its two domains, where mutual advantage asks for one competence each way. Two features of the result matter. The equality is of labels, f^- and f^+, while the domain in which each function is assessed changes between the schemes: the trading of places says that a label the old scheme attached on the strength of domain 0 is attached by the new scheme on the strength of domain 1. And the reference receives the empty list by construction, so the corollary concerns an unmarked reference and comparative descriptions; it does not establish the exchange of arbitrary, richly specified trait lists.

A worked redrawing. Take two functions, focus and organization, two domains, domestic (0) and theoretical (1), a margin $\delta = 1$, and illustrative scores, not measurements:

	focus, domain 0	focus, domain 1	organization, domain 0	organization, domain 1
branch r	7	5	8	4
branch b	4	8	5	7

The scheme Σ samples domain 0 with r as reference: b is lower on focus ($4 < 7 - 1$) and on organization ($5 < 8 - 1$), so $F_\Sigma(b) = \{\text{focus}^-, \text{organization}^-\}$ and $F_\Sigma(r) = \emptyset$. The redrawing Σ' samples domain 1 with b as reference: r is lower on focus ($5 < 8 - 1$) and on organization ($4 < 7 - 1$), so $F_{\Sigma'}(r) = \{\text{focus}^-, \text{organization}^-\} = F_\Sigma(b)$ and $F_{\Sigma'}(b) = \emptyset = F_\Sigma(r)$. Each function's comparison reverses sign between the domains, and the descriptions trade places as Corollary 1 says; whether domain 1 is a domain of focus and of organization in the sense of (A2) is the question the worked application above poses.

(S3) Balance, a conjecture.

Conjecture 2 (Balance)**.** For the reference branch and a given non-reference branch, the sets $D_r(b; \delta)$ and $U_r(b; \delta)$ carry comparable weight under a natural weighting of the competences.

This is the source's "roughly half-and-half". It needs a weighting of competences, and no weighting is natural until the inventory's granularity is settled, since splitting one competence into five multiplies its weight.

3.4 What is established and what is conjectural

(i) *Proved.* On a fixed inventory of scored competences, each of two branches can be given a non-empty list of lower traits only, by a choice of reference and inventory, exactly under δ-mutual advantage, with at most $|D_r(b;\delta)|$ lower and $|U_r(b;\delta)|$ higher traits reversing (Proposition 1); and the descriptions of a reference and another branch trade places literally, as labels and with an unmarked reference, when every function's comparison reverses sign between two sampled domains (Corollary 1). The universal form of the Principle of Symmetry over all schemes is false (Section 3.2).

(ii) *Conjectured.* Conjecture 1: for the non-autistic branch and each autistic branch, every function a current scheme describes as lower for the autistic branch has an admissible domain in which the autistic branch is higher by more than the scheme's margin. Its grounding needs, function by function, a correspondence established under (A2), of which organization is worked as an example, and a comparison measured in the corresponding domain; what counts against it is stated with it. Conjecture 2 is conjectured as well, and nothing rests on it.

(iii) *Independent of the conjecture.* The requirement of Section 3.1 does not rest on Conjecture 1: were no reversing domain found, a narrow assessment would still need an account of how its sample generalizes before supporting a general attribution, and the case for the assistance or environmental change a person wants would not depend on a compensating advantage elsewhere (Section 3.6).

(iv) *Not claimed.* That the populations have no structure of their own, or that clinical diagnoses, with their requirements of history and impairment, trade places; these are the claims (ii) and (iii) of the next subsection.

3.5 The clinic illustration, and four claims kept apart

Consider a clinic whose reference population is a branch of autistic adults and whose diagnostic manual was written from their norms, and read its intake notes on one hypothetical client from the majority.

> The client seeks sustained eye contact during speech and reports discomfort when it is not returned. Requests are seldom stated; the client encodes them as hints, expects the listener to reconstruct them, and is frustrated when the literal content is acted on. Attention moves between topics within a session, and the client reports discomfort when held to one. In group settings the client engages in repetitive rhythmic whole-body movement to music, and does not experience it as stereotyped.

Four behaviours, each stipulated of this client and none attributed to the majority as a whole, described in the register the majority's manual keeps for the minority: dancing written up as stimming; a breadth of attention as the inverse of monotropism, the attention tunnel of few interests highly aroused [20]; reliance on implicature as a difficulty with literal content. A reader who, with the heading covered, finds the notes plausible as a description of an autistic client has felt the point from the inside. The illustration is a redrawing with the reference moved and the indicators kept, and it separates the behaviour, its function for the person, its description, and whether it impairs: the register fixes the third and invites an inference to the fourth that the description alone does not establish. The source's first version, in 2024, ran the inversion on "logical" against "nuanced" (Appendix B). Four claims are then to be kept apart.

(i) Each of two branches can be given a list of lower traits only, by a change of reference and inventory. This is proved, under the stated condition.

(ii) An autistic and a non-autistic person's clinical diagnoses trade places. This needs more than (i): a diagnosis under the criteria the paper cites requires a developmental history and clinically significant impairment beside the behavioural features [10], and rewriting behaviour in clinical language does not establish that those are met. Whether those requirements are themselves reference-relative is a further argument, not made here.

(iii) The diagnostic distinction has no basis in the structure of the populations. This does not follow from (i):[1] reference-relative ranking and the structure of the population are different questions, and the second is the Branch Principle's, outside this paper (Section 10), where the source's comparison with the pre-Darwinian division of humans from animals belongs (Appendix B).

(iv) Changing the governing norms redistributes effort. This is an independent empirical hypothesis, the cost model of Section 5.

The illustration exhibits (i) and motivates (iv): if the traits are reference-relative, the cost of a mixed interaction can be sought in the relation between a person and the conventions, which a laboratory can change.

3.6 Difficulties without deficiency

Purging the deficit reading does not purge difficulty: the description, the attribution, the evaluation and the assistance or environmental change a person wants (Section 1) stay apart, and all four are kept. Someone's difficulty completing an activity they want completed is information, and it stays information under every redrawing; what the redrawing removes is the step from the description to a judgement that the person's branch is an unsuccessful version of another. The cost model of Section 5 is built to this discipline. Its competence requirement belongs to a specified protocol, not to a branch, and a shortfall against it is a relation between a person and a task; the model never carries the requirement from the protocol to the person.

3.7 Why mutual advantage is not a premise of what follows

The cost model of Section 5 predicts that a change of norms moves effort whether or not the two groups stand in δ-mutual advantage: a participant can gain from a change of norms while scoring lower on every measured competence, if the new norms presuppose less of what that participant scores lower on, and a participant who exceeds the reference somewhere gains nothing from norms that do not draw on it. The principles motivate the comparison the model makes; they are not its premises, and the model's test does not test them, which lets the empirical work proceed without settling the geometry of cognitive profiles (Section 10).

4 Context sensitivity: what toggling shows

The source makes one further observation, about a different convention: "A limitation does not toggle." An adult who switches between two modes of a convention within a day, by context, shows that "the convention is a mode the platform runs, selectable, not a ceiling the platform

[1] Two populations clustered tightly at competence profiles $(10, 1)$ and $(1, 10)$ satisfy the condition, and selecting either competence reverses which appears lower, while the populations remain sharply distinguishable types with different causes.

imposes" (Appendix B). Against the rival of a fixed limitation it is decisive, and the paper takes exactly that much from it.

Proposition 2 (What a toggle refutes)**.** *Let P be a performance and C a set of conditions. If P is observed under some $c \in C$, the claim that the person cannot produce P under any condition in C is false.*

The proposition is immediate, and its value is in what it does not say: persistence does not identify capacity, since a convention can persist across every instrument that shares its assumptions, and change does not identify convention, since a stable capacity yields different performance under different demands. The toggle is a test of *context sensitivity*: a change refutes the unconditional claim of incapacity and opens a question of mechanism, which experiments answer by varying demand, time, load and format separately from the governing norm. Applying it to differences between groups is this paper's extension.

5 The cost of interaction under governing norms

5.1 Profiles and norms

Each participant i has two profiles, both measured before any interaction.

- A *style profile* $s_i \in \mathbb{R}^p$: the participant's communicative preferences on dimensions such as directness, literalness, turn-taking and channel, scored on common scales.
- A *competence profile* $k_i \colon T \to \mathbb{R}$ over the abilities the interaction draws on, the profile of Section 3 measured for individuals.

A governing norm n, a rule set under which the interaction runs, fixes two things.

- A *presupposed style* $s_n^* \in \mathbb{R}^p$: the communicator the rules are written for, coded from the rules' text by raters who do not see the outcomes.
- A *competence requirement* $\theta_n \colon T \to \mathbb{R}$ with weights $w_n \colon T \to [0,1]$: how much of each competence the rules call on, and how heavily.

5.2 The cost model

Hypothesis 1 (The cost model)**.** Participant i's effort under norm n is

$$c_i(n) \;=\; a_i \;+\; m_i(n) \;+\; \varepsilon_i(n), \qquad m_i(n) \;=\; \lambda_s \, d\big(s_i, s_n^*\big) \;+\; \lambda_k \sum_{\tau \in T} w_n(\tau)\, \big[\theta_n(\tau) - k_i(\tau)\big]_+ , \quad (2)$$

with $[u]_+ = \max(0, u)$, d a distance on style profiles, $\lambda_s, \lambda_k \geq 0$, a_i a participant-specific baseline over the conditions studied and $\varepsilon_i(n)$ zero-mean noise.

The two terms of the mismatch $m_i(n)$ have different shapes, and the difference is the point.

- *The style term is symmetric.* Departing from the presupposed style in either direction costs effort: a direct communicator under indirect rules and an indirect one under direct rules both adapt. A distance is the right form for a mismatch of styles, where neither pole is better.

- *The competence term is one-sided.* Falling short of a requirement costs effort; exceeding it costs nothing. A participant whose competence exceeds what the rules call on is not penalized for the excess, which a symmetric distance on competences would wrongly do.
- *The baseline is a baseline.* a_i collects whatever is constant across the conditions studied: capacity, in part, but also sensory load, fatigue and anything else the conditions share. It is not identified as capacity, and the model does not call it that.

Status. Equation (2) is a hypothesis about effort, which the study tests. Reference-relativity, (S1), makes the comparison between a participant and a norm well defined, the norm's requirement being the reference for the one-sided term, and entails nothing about how effort tracks it.

5.3 The predictions, exactly

Write $G(n)$ for the expected group gap in effort under norm n, the minority group's mean less the majority group's,

$$G(n) \;=\; \Delta a \;+\; \Delta M(n), \tag{3}$$

with Δa the baseline gap and $\Delta M(n)$ the gap in mean mismatch, both signed in the same direction.

P1, the gap under the default norms n_0. $G(n_0) = \Delta a + \Delta M(n_0)$. Under norms that presuppose the majority's style and draw on its competences, $\Delta M(n_0) > 0$ whenever the minority's profiles lie farther from $s^*_{n_0}$ and fall shorter of θ_{n_0}, and the gap is the sum of two parts that the default condition cannot separate.

P2, norm sensitivity, the primary prediction. For a second norm n_1,

$$G(n_1) - G(n_0) \;=\; \Delta M(n_1) - \Delta M(n_0), \tag{4}$$

whatever Δa is; and within each person the baseline cancels,

$$c_i(n_1) - c_i(n_0) \;=\; m_i(n_1) - m_i(n_0) \;+\; \varepsilon_i(n_1) - \varepsilon_i(n_0), \tag{5}$$

so that a participant's change in effort is predicted, up to noise, by the change in a quantity measured before the interaction. Equation (5) is the strongest test the model offers, because it runs on premeasured profiles, person by person, and because the restrictions M_γ and M_μ of Section 6 predict a slope of zero for it.

P3, reversal, the strong form.

Proposition 3 (The reversal condition)**.** *Suppose $G(n_0) > 0$. The gap reverses under n_1, $G(n_1) < 0$, exactly when either of the following equivalent conditions holds:*
(a) the norm-induced reduction exceeds the initial total gap, $\Delta M(n_0) - \Delta M(n_1) > G(n_0)$;
(b) under the new norms the mismatch term favours the minority by more than the baseline gap, $-\Delta M(n_1) > \Delta a$.

Proof. By (3), $G(n_1) = G(n_0) - [\Delta M(n_0) - \Delta M(n_1)]$, which is negative exactly under (a); and $G(n_1) = \Delta a + \Delta M(n_1)$, which is negative exactly under (b). □

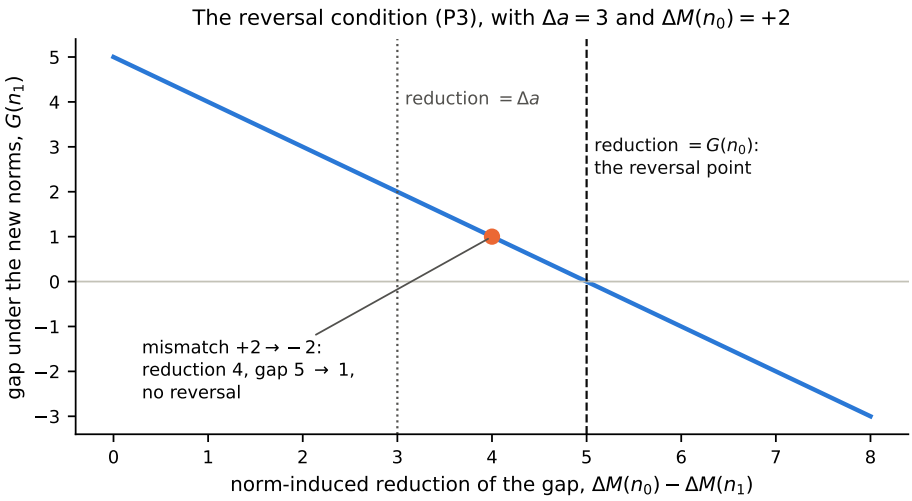


Figure 1: The reversal condition of Proposition 3. The gap under the new norms falls linearly in the norm-induced reduction; it closes when the reduction equals the whole initial gap $G(n_0)$, not when it equals the baseline gap Δa. The marked point is the worked case in the text: a mismatch contribution moving from $+2$ to -2 reduces the gap by 4, more than the baseline gap of 3, and the gap falls from 5 to 1 without reversing.

The condition is stricter than the natural reading of "the norm effect outweighs the baseline". Take $\Delta a = 3$ and let the mismatch contribution move from $+2$ under n_0 to -2 under n_1. The reduction is 4, larger than the baseline gap, but smaller than $G(n_0) = 5$: the gap falls from 5 to 1 and does not reverse. Reversal needs a reduction greater than 5, equivalently a new mismatch contribution below -3; Figure 1 draws the arithmetic. In the language of aptitude-by-treatment interaction, P2 is an interaction of group with condition, and P3 is the condition for it to be disordinal, the groups' lines crossing between n_0 and n_1.

6 The model family, and the test that separates its members

6.1 One model, four restrictions

Add to (2) the two terms a rival would supply: a condition effect γ_n common to all participants, and a mismatch with the partner. For participant i with partner j under norm n,

$$c_i(n, j) \;=\; a_i \;+\; \gamma_n \;+\; m_i(n) \;+\; \mu\, d\big(s_i, s_j\big) \;+\; \varepsilon_i(n). \tag{6}$$

The models compared are restrictions of (6), and each makes a definite prediction for a change of rule set with the partners held fixed (Table 2).

No tradition is identified with a restriction. An account that emphasizes individual limitations can predict effects of task demands and of accommodations, and so is not M_γ or M_μ by fiat; a double-empathy account can include norm effects, and so is not M_μ by fiat. The restrictions are compared as models, by what they predict for the data of Section 7.

Table 2: The model family: four restrictions of (6), and what each predicts when the governing rule set changes and the partners stay the same.

Model	Restriction	Prediction under a change of rule set, partners fixed
M_γ, general protocol benefit	$\lambda_s = \lambda_k = \mu = 0$	Effort moves by γ for everyone; nothing is redistributed, and nothing is moderated by premeasured mismatch.
M_μ, partner mismatch	$\lambda_s = \lambda_k = 0$	Beyond γ, nothing changes, since the partner is unchanged.
M_λ, norm mismatch	$\mu = 0$	Effort shifts toward whoever mismatches the governing rules, in proportion to premeasured mismatch (P2), and reverses only under the condition of Proposition 3.
M_{full}	none	Both the partner and the norm terms.

6.2 The double-empathy model, read from its text

The double-empathy literature's formal model is the distinct communication strategies model [8], read here from its text. Its implemented mechanism is channel preference: each agent registers the partner's empathy as a weighted average of the partner's verbal and nonverbal output, the verbal weight the agent's own preference *cp*, and the gap between expected and registered empathy raises defensiveness, which lowers output until the loop can collapse. Its text states that expectations "may depend on context, relationship history, communication norms" while holding expected empathy fixed in its simulations, and that "the costs are therefore bilateral but need not be equal or arise from the same source"; it has no parameter for a governing norm. It is therefore silent on norms as implemented and open to them in its text, and the distinction between it and the cost model can be shown in data only under assumptions, stated next.

The mediation analysis, and what it can establish. Suppose the double-empathy mechanism is extended so that expected empathy depends on the governing norm. A change of rule set then acts on effort *through* the gap between the empathy a participant expects and the empathy the participant perceives: the gap moves, and effort moves with it. Under M_λ effort moves with premeasured profile-to-norm mismatch whether or not the gap moves. Three things follow for the design, and together they are less than a test.

(i) *The mediator is the gap, not perceived empathy.* Expected empathy may change with the norm while perceived empathy does not, and the whole change in effort may then run through the gap; an analysis that controlled perceived empathy alone would count it as a norm effect beside the mechanism.[2] The design therefore measures both, expected partner empathy before each condition's task and perceived after it, and the mediator is their difference.

(ii) *Controlling the mediator does not identify the direct effect by design.* The rule set is randomized; the gap is not. The coefficient of premeasured mismatch with the gap's change controlled is the direct effect only under sequential ignorability [21, 22], which measurement

[2]Under rule set A a participant expects 8 and perceives 6; under B expects 6 and perceives 6: the gap falls from 2 to 0 with perceived empathy unmoved.

error in the reports, common causes of the reports and of effort, and feedback from effort to the judgement of the partner each violate. The analysis is therefore assumption-dependent: direct and indirect effects under sequential ignorability, the sensitivity analysis of Imai, Keele and Yamamoto [22] over a range of confounding fixed in advance on substantive grounds, and measurement error carried with the confounding by the joint sensitivity analysis of Liu and Wang [23], the reports' reliabilities from the pilot being its inputs and not a correction on their own.

(iii) *The two accounts are not exclusive.* Equation (2) does not say by what route mismatch produces effort, and a positive λ may describe an effect that runs in part through the gap. A direct effect that shrinks when the gap is controlled is consistent with the cost model; no categorical prediction, zero, the full slope or something between, is made for the residual coefficient; and a direct effect that vanishes at a stated precision would show that the first study cannot find a norm effect beside the gap, not that none exists.

What the first study can establish is whether the mismatch effect is present with the gap controlled, and how much of it the gap carries under the stated assumptions and across the stated sensitivity range. What would separate the mechanisms is a study that manipulates expected empathy directly, by disclosure or by instruction, which is deferred (Section 7).

7 Study 1: a discriminating design

The design manipulates the governing rules, holds fixed what can be held fixed of everything else, and measures what cannot. A comparison of "default conversation" with "an explicit protocol" would change norms, structure, predictability, memory demands and perhaps modality at once, and an improvement under the protocol could be any of them. The study therefore compares two explicit rule sets.

Participants. Mixed dyads, one autistic and one non-autistic adult, with diagnosis by report and the autistic participants' self-identification recorded.

Before any interaction.

(i) Each participant's style profile s_i and competence profile k_i are measured with instruments fixed in the preregistration [24]: for style, self-report and a brief performance measure of directness, literalness and turn-yielding preference; for competence, a task of implicature comprehension and a task of following explicit structure, each scored on the scale the requirement is written in.

(ii) Independent raters, blind to the outcomes, code each rule set's presupposed style s_n^* and competence requirement θ_n from its text, on the same scales.

(iii) The distance d, the weights w_n, the outcome measures and the analysis are fixed in the preregistration. Profiles, metric and norm codings chosen after the outcomes are seen would make the model too flexible to fail.

(iv) The common metric is established, not assumed. Independent coding protects the norm locations from being chosen to fit the outcomes; it does not make a participant's preference score and a coded norm location commensurable. The pilot therefore codes the rule sets on

the anchored scales the profile instruments report on, infers the competence thresholds from the rule text on the scale the competence tasks are scored in, and checks the calibration against behaviour: participants' adherence to each rule set and their reported difficulty with it should order with the coded distance, and where they do not the scales are revised before the main study.

A worked instance. The notation carries practical choices, and one pair makes them visible. Let the style instrument report directness and literalness on anchored seven-point scales (directness 1: requests made only by hint; 7: requests always stated outright), and let the raters place rule set A at $s^*_{\mathrm{A}} = (1.5, 2)$ and rule set B at $s^*_{\mathrm{B}} = (6.5, 6)$ on the same anchors, from the rules' text. Let the implicature task and the explicit-structure task each be scored 0 to 10, and let the raters set rule set A's requirement at $\theta_{\mathrm{A}} = 6$ on implicature, the level of indirectness its hints demand, calibrated in the pilot against the task's items, and rule set B's at $\theta_{\mathrm{B}} = 5$ on explicit structure, each with weight one on its own competence. An autistic participant measured at $s = (6, 6)$, implicature 4, structure 8, has style distance 6.0 to A and 0.5 to B and a shortfall of 2 under A and 0 under B; the non-autistic partner at $s = (2.5, 3)$, implicature 8, structure 5, has distances 1.4 and 5.0 and no shortfall under either. With $\lambda_s = 0.6$ and $\lambda_k = 0.8$ in workload-score units per scale unit, values the study estimates and the simulation only assumes, the first participant's mismatch is 5.2 under A and 0.3 under B, a predicted change of -4.9 from A to B, and the partner's is 0.85 and 3.0, a change of $+2.15$: the pair's predicted points on the P2 line. Every number in the chain is a choice the pilot has to make good: that a 6 on a rule's requirement means what a 6 on the task means, that the raters' 1.5 and the participant's 2.5 are a unit apart in the sense the distance assumes, and that the Euclidean distance is the right d for two dimensions that need not count equally.

The manipulation. Two explicit rule sets differ in the communicator they presuppose and are matched on specified incidental features: length, number of rules, channel, time, and the structure of the task they govern. They are not matched in everything, and the model does not ask them to be. The competence term says that the rule sets draw on different competences; inferring a hint and yielding a turn on an implicit cue make demands that stating a request and yielding in words do not, and those are the differences the model is about. What remains, the learning burden of each rule set and its automaticity after practice, is measured rather than assumed equal: training time to criterion, practice errors and reported effort during practice are recorded for each rule set and serve as the check on the matching, which is treated as a claim the pilot supports or revises. They do not enter the primary analysis as covariates: measured after the participant has met the assigned rules, they can carry part of the adaptation cost the analysis estimates, and adjusting for them would remove that pathway or bias the estimate, the hazard of conditioning on a variable the manipulation affects [25]. The primary estimand is the total condition-dependent mismatch effect; how training burden relates to effort is a separate analysis with its own estimand, stated in the preregistration.

- *Rule set A* encodes indirect, implicature-rich conventions: requests made by hinting, an opening exchange of small talk, turns yielded by implicit cues.
- *Rule set B* encodes direct, literal conventions: requests stated outright, no obligatory small talk, turns yielded in words.

Both are practised to a criterion before the task; adherence to the rules, coded from recordings, is the manipulation check. Which conventions rule set B should encode, and from what sources,

is a decision for the preregistration, informed by autistic participants' own accounts of their communication and by the published descriptions of autistic communication preferences; the design does not presume that rule set B is the autistic participants' preferred style, it measures where each participant stands relative to each rule set.

Within dyads. Each pair works under both rule sets, in counterbalanced order, on two parallel versions of a collaborative task, also counterbalanced. The order is assigned to dyads at random and balanced; a period effect is estimated as the order-by-condition term; differential carryover, which a two-period crossover cannot separate from the order effect [26], is guarded against by the parallel task versions and a washout between conditions and is assessed by the adherence coding of the second condition, in which the first rule set's conventions would show if they persisted. The room, lighting and noise are identical across conditions; fatigue is measured after each condition. The pair is its own control: the baseline a_i, the partner term $\mu\, d(s_i, s_j)$ and the dyad's shared conditions cancel in the within-person difference (5).

Outcomes. The primary outcome is one measure, fixed in advance: the subjective workload score, the mental-demand and effort items of the NASA Task Load Index [27] reworded to the conversation just held (how mentally demanding the conversation was; how hard the participant had to work to follow the rules and accomplish the task), on the index's scale, per participant and condition. A behavioural index, the rate of repair initiations and restarts per minute of the participant's speech coded from the recording, is a secondary measure of convergent evidence, not a second primary, so the testing rule has one outcome. Its relation to effort is not assumed, since clarifying questions can be an efficient strategy and effortful speech can be fluent: the pilot establishes what the rate captures and whether its per-minute denominator stays comparable when the protocols change speaking time. Its raters are blind to the hypotheses, to group, profiles and condition labels, though not to the conditions themselves, which are audible. Low reported effort can mean easier coordination or lower engagement, so effort is read with task performance and adherence beside it, and a condition in which all three fall is read as consistent with disengagement rather than with ease. Task performance and rapport, secondary; expected and perceived partner empathy per condition, whose difference is the mediator; rule adherence, the manipulation check; training time, practice errors and practice effort per rule set, the matching check.

Predictions.

- *Primary (P2).* On effort, the rule set interacts with premeasured mismatch within persons: the change in effort from A to B is predicted by the change in m_i, with slope one in the model's units and positive in any units. The group-level redistribution of effort is this prediction's aggregate.
- M_γ. A main effect of rule set only, with no moderation by premeasured mismatch.
- M_μ. No change beyond the main effect, the partner being unchanged.
- *A norm-dependent double-empathy mechanism.* A change carried by the expected–perceived empathy gap. What the first study can say about it is stated in Section 6; it is not a prediction the primary analysis tests.

Analysis. The primary analysis is the within-person difference regression of (5), the change in the workload score on the change in premeasured style mismatch and competence shortfall, with standard errors clustered by dyad. The preregistered decision rule for *primary success* has three parts: the joint Wald test of λ_s and λ_k rejects at 5%; at least one coefficient is positive by its one-sided 5% test; and neither coefficient contradicts its sign restriction, the one-sided 5% test of $\lambda \geq 0$ rejecting. The first part asks whether premeasured mismatch moderates the change in effort at all, the other two whether it does so in the model's direction with the uncertainty of each estimate respected, since the model permits a coefficient of exactly zero (Section 8 reports the rule at that boundary). Because success requires the joint rejection, the rule's false-positive rate under the joint null cannot exceed the joint test's. Joint rejection with a sign contradiction is recorded as moderation against the model's direction and counts against the cost model as stated; joint rejection with neither coefficient positive by its one-sided test is recorded as moderation of undetermined direction; neither is success. Evidence for each coefficient separately is secondary, tested one-sided in the predicted direction with the familywise rate held at 5%. Levels are analysed in the actor–partner interdependence model [28], participants nested in dyads and conditions within participants, which identifies μ from between-dyad variation only under an assumption the design does not supply, that the baselines have zero conditional mean given the levels regression's regressors, the measured profile coordinates included as covariates centred within group; with one partner per participant, $a_i' = a_i + \mu\, d(s_i, s_j)$ with $\mu' = 0$ gives the same observations as a_i with μ, so the assumption is what identifies μ and the supplement shows what its failure does. The four models are compared by their Bayesian information criterion on the exact likelihood stated in the supplement, and the mediation analysis is as stated in Section 6. Appendix A lists what the preregistration fixes.

What would count against the model. Stated in advance, with the preregistration, and each with the precision it needs, since a nonsignificant estimate establishes no absence:

(i) a manipulation that changes adherence but moves no one's effort, the condition effect and both mismatch coefficients bounded within their smallest effects of interest;

(ii) a change of effort that premeasured mismatch does not moderate: the joint test not rejecting, and an equivalence test [29, 30] rejecting, for each coefficient, every value beyond its smallest effect of interest; a nonsignificant moderation whose interval still covers the smallest effect of interest is inconclusive, not a disconfirmation;

(iii) a direct effect of premeasured mismatch bounded within the smallest effect of interest once the change in the empathy gap is controlled, with the sensitivity analysis showing the bound robust across the prespecified range of mediator–outcome confounding.

None of these falsifies the model family, whose parameter space contains $\lambda_s = \lambda_k = 0$; the first two establish that mismatch effects of the prespecified size are absent for the studied participants and protocols. The smallest effects of interest are provisionally a quarter of the simulated values, 0.15 and 0.20, a sensitivity setting to be translated, on the fixed scales, into the smallest change in workload per unit of mismatch that would matter; power to detect them and power to establish their absence are reported separately (Section 8). The third condition would send the question to the study that manipulates expectations.

Deferred. Three manipulations go to studies of their own: composition, same-neurotype dyads beside mixed ones, testing whether the same-neurotype advantage of the first diffusion-chain study [31], absent in its replication [32], appears under a change of norm; disclosure of diagnosis, which manipulates expectations directly and is the study the mediation analysis defers to; and a round

robin [33], which identifies the partner term within persons. Crossing norms with any of them in a first study would multiply the dyads and confound the manipulation checks.

8 The simulation, in summary

Before any data, the design was run on simulated participants; the supplement reports the full study, from the exact likelihood to a robustness study of seven scenarios, and its script, `norm_cost_sim.py` version 1.5, runs with a fixed seed. The settings are 60 mixed dyads under two rule sets presupposing opposite styles and drawing on different competences, two-dimensional style and competence profiles drawn around overlapping group means, a baseline gap of 0.5 against a norm-induced gap near 1.5 in either direction, and a dyad effect shared by the two partners; the generating values are $\lambda_s = 0.6$, $\lambda_k = 0.8$, $\mu = 0.4$ and $\gamma = -0.3$, assumptions chosen for illustration and not estimates.

What the simulation shows, at those settings, is this.

- *Recovery and power.* Every parameter is recovered without bias and with coverage near nominal; the within-person coefficients are estimated to a tenth of their values, the condition effect has power 81% and the partner term 76% at 60 dyads. The decision rule succeeds in every study at the generating effects and, at a quarter of them with both coefficients small, in 98.5% of studies at 60 dyads (94% with the profiles and codings measured with error), which is the figure the preregistration should quote for the two-coefficient alternative; with one coefficient at its value and the other exactly zero the rule succeeds in 95% and 94%, and with one small coefficient alone in 66% and 57%, so that a small effect confined to one component needs a larger study. The rule's false-positive rate when no mismatch effect exists is 1.9%. The equivalence tests at the smallest effects of interest reach 80% power only above 100 dyads: a study sized to detect the effects at 60 dyads is not sized to rule them out (Figure 2).

- *Model recovery.* The Bayesian information criterion on the exact likelihood selects the generating model in 97%, 76%, 96% and 79% of studies for the four models, its errors almost all the dropping of the partner term; no study generated with the norm terms selected a model without them. A dyad-held-out predictive score over-selects the larger models when the truth is small, the known behaviour of such scores [34, 35], and the criterion is the preregistered comparison.

- *A rule set the model has not seen.* Fitted to rule sets A and B on five sixths of the dyads, the mismatch coefficients predict the relative individual effects of a withheld rule set C for the held-out dyads, from their premeasured profiles alone, to the noise floor (1.10 against the rivals' 1.47, floor 1.10).

- *The signature.* Figure 3 shows what the study's main outcome looks like under norm mismatch and under a general protocol benefit.

- *Robustness.* The joint test's power, the criterion's selection of a norm model and the within-person estimates survive measurement error in the profiles and codings, halved effects, correlated baselines, unequal residual variances and correlated predictors, singly and together. Two things do not: a rule set's coding error, one draw shared by every participant, spoils the intervals of the condition effect and the competence coefficient, so that the codings must be made by several raters with their disagreement propagated; and baselines correlated with partner distance bias the partner term by more than its value, as the identification argument of Section 7 says they must, the profile covariates absorbing almost none of it.

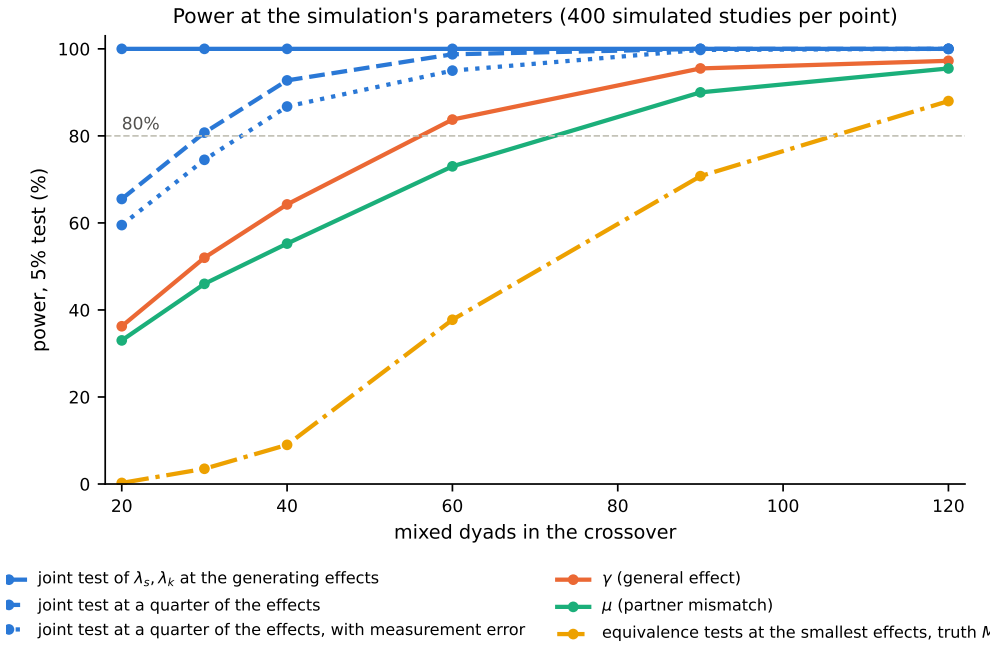


Figure 2: Power against the number of mixed dyads, 400 simulated studies per point. Solid lines: the joint test of (λ_s, λ_k) at the generating effects, and the two-sided tests of the condition effect γ and the partner term μ. Dashed: the joint test at a quarter of the mismatch effects ($\lambda_s = 0.15$, $\lambda_k = 0.20$); dotted: the same with the profiles and the rule-set codings measured with error; the decision rule's power runs within six points of the joint test's at 20 dyads, within three at 30 and within two from 40. Dash-dotted: the power of the two equivalence tests at the smallest effects of interest when the truth is M_γ, which passes 80% only above 100 dyads.

The simulation does not show that the effect sizes are realistic or that the profiles and codings are commensurate, which is the pilot's problem, nor anything about the empathy gap, whose analysis rests on assumptions a simulation that built them in could not test.

9 What existing results constrain

Three bodies of results bear on the design.

Information transfer in chains. The first diffusion-chain study found that information passed along same-neurotype chains more effectively than along mixed chains [31]; the registered replication with 311 participants found no main effect of chain type on transfer, for fictional or factual stories, with an unpredicted steeper fall along autistic chains for fictional ones, and rapport effects that depended on role and on being told the chain's neurotype [32]. The constraint is twofold: information transfer is not where the cost appeared in that task, and the rapport effects depend on role, which is why effort rather than rapport is the primary outcome here.

Rapport in groups. In 36 groups of four, all-autistic, all-non-autistic and three-to-one majorities, there was no overall group-type difference in composite rapport [36]; all-autistic groups rated their

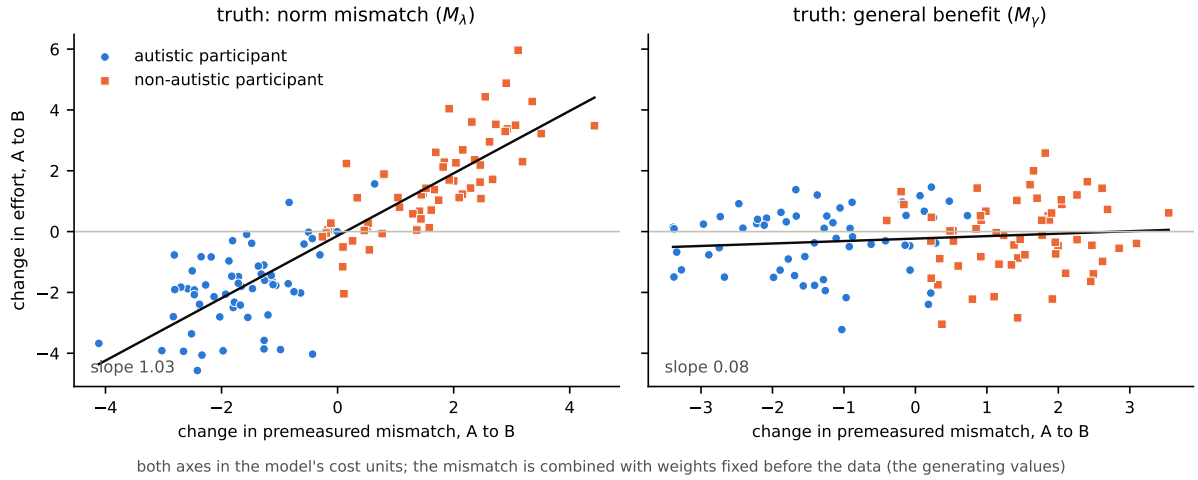


Figure 3: The P2 signature in one simulated study of 60 dyads under each of two truths. Each point is a participant: the change in effort from rule set A to rule set B against the change in premeasured mismatch, combined with weights fixed before the data, in the model's cost units. Under norm mismatch (left) the change lies on a line of slope near one and the groups move in opposite directions; under a general protocol benefit (right) the slope is near zero and the groups move together.

interaction more enjoyable and friendlier than both kinds of mixed group, and autistic participants' ease fell as non-autistic members were added, in item-level results their authors call preliminary. The study did not manipulate norms, so a reading of that pattern through the cost model, the autistic participants' ease falling as the session's conventions shift toward the majority's, is a hypothesis for Study 1 and not a result.

Perception. Non-autistic observers form less favourable first impressions of autistic adults from brief samples, on the basis of presentation rather than content [37], which is one reason the cost of a mixed interaction cannot be read off its outcome alone.

10 Outside this paper

Three things the source raises are extensions, and the paper says of each what it would take.

Branch geometry. The Branch Principle is a hypothesis about the structure of cognitive profiles: at matched resolution, non-autistic profiles form one type, or few, and autistic profiles form several, in the strong form about as distant from each other as from the non-autistic type. Its test fits the same latent-structure models to both groups on the same cognitive-profile measures, a mixture model, an additive tree [38] and a dimensional model, and compares them by likelihood; the decomposition of autism's phenotypic heterogeneity into classes with distinct genetic programs [39] is evidence at a different grain. The geometry is independent of the cost model and is not support for mutual advantage, since equal distances do not imply crossed advantages ($b = (1, 1)$ and $c = (1, -1)$ are equidistant from $r = (0, 0)$; b dominates r, c and r cross), and linking it to Proposition 1's premise needs an argument of its own, for which an asymmetric similarity [40]

may be the instrument.

Linguistic masking. The source names a phenomenon a layer below behavioural camouflaging [41, 42, 43]: a speaker who learns a cognitive term's public extension case by case over a private intension that differs from the dominant group's, so that the difference shows only on new cases. Its signature, a generalization gradient, is not unique to it, and what would tie it to the mechanism is a private organization of the instances, measured by similarity sorting without the term, predicting the divergence over and above familiarity and confidence; which instruments it would reach is open, since an eight-item alexithymia score shows practically ignorable differential item functioning between the groups [44]. This is a second paper.

Shared absence. A further hypothesis, that non-reference branches share the absence of the reference's conventions and so find each other's interactions cheaper even where their own preferences differ, is stated with its premises when it is tested, against familiarity, stigma, preference variability and disclosure; it predicts nothing reported here.

11 Discussion

The classification argument. The paper began from one person's encounter with three questions and asked what licenses a general attribution of a function from performance in a sampled domain. Its answer is a requirement, not a verdict: an attribution earns its generality through an account of how the sampled manifestations generalize, and where the account is missing the conclusion has the scope of its sample. The requirement stands whether or not Conjecture 1 does. The apparatus that makes the requirement precise, classification schemes with their sampled domains, indicator rules and designated reference, and the admissibility conditions on redrawing them, is the paper's first contribution, and it turns the Principle of Symmetry from a slogan into a claim with a proof where it can have one and a stated criterion of evidence against it where it cannot (Section 3.4). The decisive missing pieces are the correspondences and the comparisons in the corresponding domains: until it is shown, function by function, that activities in a second domain are manifestations of the function the first domain samples, and the branches are compared there, the conjecture is a programme, and the organization example shows the form its arguments must take, not that they succeed.

The cost model. The second contribution places part of a mixed interaction's difficulty in the rules and makes the placement measurable: the share of the group gap that moves when the rules move, person by person, from profiles measured in advance. The model family keeps the estimate honest, since a general benefit of explicit rules and a mismatch with the partner are both allowed to carry the data if they can; the reversal condition is a caution to the design, since a study powered for the reduction of a gap is not powered for its crossing.

Limitations.

- *Measurement.* The cost model is only as good as the commensurability of the profiles and the norm codings, which fixing them in advance does not secure; the pilot's calibration is the remedy, and the supplement's robustness study shows what measurement error costs.

- *Two rule sets.* Two norms identify the norm terms but not their functional form; a third rule set is the natural extension and the strongest test.
- *Dyads.* The partner term is identified between dyads, needs 90 or more pairs at the simulated effect size, and rests on the zero-conditional-mean assumption of Section 7; a round-robin design identifies it within persons.
- *Scope.* Both conditions impose explicit instructions, so what the first study measures is adaptation to experimental protocols; extending it to the norms of ordinary interaction needs evidence that the protocols represent them.
- *Participation.* The domains, the correspondences, the rule sets, the profile instruments and the interpretation of effort should be developed with autistic researchers and participants, not only about them; the choices left open are left open for that reason.

Two research priorities. The paper's two parts call for two programmes, and neither substitutes for the other. The first establishes and tests cross-domain correspondences for the functions current schemes attribute, beginning with organization, the worked candidate, and with White, Burgess and Hill's finding [19], which favours the conjecture only on its implicit-demands reading, others' demands against the reasoner's own, and otherwise predicts an autistic disadvantage in open-ended theoretical work as well: tasks with matched dependency structure in a practical and a theoretical setting, the correspondence weighed on the evidence of Section 3.2, and then the comparison of branches in each admissible pair of domains, where a specified study can support Conjecture 1 or count against it, and can decisively test its restriction to the study's prespecified domains. The second tests how governing communication norms redistribute effort: Study 1, preregistered, and its extension to a third rule set. The crossover addresses the second and does not settle the first. For the extensions, a cognitive-profile data set at matched resolution is what would let the branch geometry be fitted rather than argued.

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A Preregistration items

The preregistration fixes, before any participant is run: the instruments for s_i and k_i and their scales; the two rule sets' texts, their coded s_n^*, θ_n and w_n, and the rater protocol; the distance d; the effort measures, subjective and behavioural, and the expected and perceived empathy reports; the practice-to-criterion rule and the adherence coding; the counterbalancing; the primary outcome, the workload score, with the repair index secondary; the primary analysis of (5) with dyad-clustered errors, the three-part decision rule of Section 7 and how rejections with a sign contradiction or of undetermined direction are recorded, the secondary one-sided tests, and the smallest effects of interest on the fixed scales; the period and carryover terms; the separate analyses of training burden, with their estimands; the actor–partner model for levels with the group-centred profile covariates and its zero-conditional-mean assumption; the model comparison by the Bayesian information criterion on the exact likelihood; the mediation analysis with its sensitivity analysis; the three failure conditions of Section 7; and the sample size, with the intended alternative named: 60 mixed dyads for the two-coefficient alternative at a quarter of the simulated effects, more if a small effect confined to one component is the alternative of interest (Section 8), 90 if the full family is to be compared, and 120 if the equivalence tests at the smallest effects of interest are to have 80% power or more, at the simulated effect sizes, which are to be replaced by pilot estimates where available. Before the preregistration, a pilot establishes whether the profiles and the norm requirements can be scored on shared scales, whether participants can follow both rule sets to criterion and at what cost in training, whether the effort measures track the intended construct against performance and adherence, and the reliabilities of the empathy reports and of the rule-set codings.

B The source passages

The passages below are quoted as written in the larger work [1], from the section "The Aethic Mindset and Autistic Savants" and its subsection "Illustrating the Bias of Perspective in Diagnosing Logic", and from the paragraph "The decisive fingerprint: it toggles" of the section "Aethic Reasoning Against the Modal Structure of Human Language". The source dates the two principles to a short video the author recorded on June 8, 2024, at 2:39 PM ET.

The Branch Principle, and its motivation.

> "Ontologically neurodivergent people are better understood as a collection of categorical branches and 'allistic' is a single branch, such that two random branches are typically as different from one another as they each are to a third."

> "A better model for understanding autistic minds than the autistic-allistic binary is more of a tree model, such that allistic people are one branch alone and the rest have been all categorized as 'autistic'. Inadvertently speaking, then, this proposes that an autism diagnosis is not actually diagnosing a person's traits to the level of specificity of an allistic person (as would assign them a particular branch) but instead broadly classifies them as 'non-allistic' for their lacking allistic traits. In reality, though, each branch comes with a different set of cognitive intuitions, which are separate enough so that two autistic people drawn from two different branches tend to be as different from one another at the cognitive level as either of them are from allistic people."

> "This is meant to be directly analogous to the post-Darwinian tree of life model treating humans as a branch, as opposed to the pre-Darwinian model in which humans and animals are two categories."

The Principle of Symmetry, and its intended content.

> "You can always redraw the conventional category boundaries in psychology so that allistic and a given neurodivergent branch 'trade places phenomenologically', in the sense that the exact behavioral diagnosis, as an Aethus, is made to reverse isomorphically."

"As an Aethus" is glossed at the source as "a partial information state about that behavioral diagnosis rather than an exhaustive description". The principle's intended content is given in the same passage with the author's own case as the instance, offered in this paper as motivation only:

> "What's fundamentally wrong about this is that it fundamentally misunderstands what's actually going on: what's going on is utter, complete symmetry between cognitive branches. That is, for allistic people versus a given branch of autistic people, there will be roughly half-and-half instances across traits for which the autistic person is infantile versus for which the allistic person is infantile. Using myself as an example, at age twenty I still had the emotional maturity of a thirteen year old, but at age thirteen I already had the mathematical maturity of at least a twenty year old. Had my branch been the reference instead, allistic people would look just as profoundly infantile in so many different ways to how I looked in social, emotional, and life skills."

The random-partition illustration.

> "This is also where categorizing all the non-allistic branches as one 'autism' is measurably problematic: you can find an autistic branch for which any trait is infantile, thereby seeming to confirm your prior that 'look, autistic people are infantile in this way'. Running the same logic on a random partition where we put half the branches, including the allistic one, in one category while the rest are placed in the other, and we could just as easily misinterpret allistic people as infantile in any conceivable way by finding it in another branch of the categorization and then mistakenly importing it to an assessment of allistic people."

The 2024 version of the thought experiment, on "logical" and "nuanced". The passage that follows the two principles:

> "The observation, now, is in how mathematics itself, especially in an educational setting, is commonly called 'logical' in the sense of there only being right and wrong answers, whereas the humanities are instead called 'nuanced' in the sense that they resist binary 'rightness' of answers. What I want to point out now is that this take itself is actually biased to the allistic perspective, which has some intriguing philosophical implications once we start to unpack it. First of all, notice how due to the second bullet point, it is actually universal that anyone learning the most introductory material to a topic of a different intuitive landscape will report the same characterization. In order for a child to learn literature, they must first start by learning the alphabet and the sounds they make. In this sense, like twelfth grade math, kindergarten literature has clear rules—either the letter is written this way or it is not, and either the letter makes this sound or it does not. The fact that kindergarten literature is at an isomorphic introductory level as twelfth grade mathematics is then simply an illustration of how the general population—most of them being of the allistic cognitive branch—hold a vastly higher intuitive capacity for literature over mathematics. So in this sense comparing twelfth grade mathematics to twelfth grade literature is the wrong reference class, hence contributing to this bias of perspective that literature is inherently more nuanced than math."

The executive-functioning case and the six traits. From the author's note of October 24, 2026, 1:48 PM ET, recorded in the larger work beside the Principle of Symmetry:

> "I myself was talking to a psychiatrist the other day and she asked me the question of 'how is your focus, organization, and initiation' for 'executive functioning', and basically the understanding is that I'm supposed to say 'not great' because in a real-world setting I'm late with the laundry a lot and things like that. Thing is, ontologically what I suspect happened is that you can picture all my executive functioning as having been picked up and moved into straight up inductive reasoning for theory-building and things like that, in which sense all three traits are fantastic, and I'm literally holding a huge network of 'good move'-considerations in mind at all times. [...] Basically because of where my executive functioning is situated it falls totally under the radar to standard measures and gets diagnosed as 'not there', and the principle of symmetry then specifically says for this that there exists a new way to draw the category boundaries such that the exact allistic and me branches' own diagnoses flip on the matter, or even to arbitrary precision over several matters like as an Aethus augmentation. [...] Literally you name it—pick three positive-existing traits for allistic under executive functioning, and three negative-existing traits for my branch under executive functioning (hence notice how they're just six total traits, not granted 'deficient' or not, which would be asinine in terms of killing the actual relativism at play by imposing mainstream-bias on it), and you can literally assign these same traits respectively to any two branches you pick, including the swap case of those very same two! (Not to be confused though with assigning those three traits to respectively allistic and autistic, since my branch principle has the association with branch-swapping as a category error—instead the isomorphism demands you swap them to 'single branch versus multiple (specifically remaining collection of) branches' kind of thing)."

The seat of the cost. From a footnote of the same section:

> "[N]otice how many autistic people themselves are described as taking a script to allistic conversations. Once again, notice how if the shoe were on the other foot, for any given non-allistic cognition branch allistic people practicing its topics would be interpretable as using a script."

The toggle.

> "One further fingerprint is of a different kind, and it is decisive against the strongest rival reading: that the convention is not a convention at all but a fixed cognitive limitation. A limitation does not toggle. The switching just described—an adult moving, on demand, between the attached picture and the conventional one within a single day, by context—is a *dissociability* result: the convention is a mode the platform runs, selectable, not a ceiling the platform imposes."